
SUMMARY

Machine Learning Engineer II at Amazon building AI/ML platform systems in Python, Java, TypeScript, Spark, and AWS. Recent work covers agent lifecycle infrastructure, evaluation, deployment guardrails, monitoring, data pipelines, and SageMaker training workflows.

SKILLS

- **Programming:** Python, Java, TypeScript, SQL
- **AI/ML Platforms:** Agentic Systems, MCP, Agent Evaluation, LLM-as-Judge, MLOps, Causal Inference, EconML, SageMaker
- **Data & Infrastructure:** Spark, AWS CDK, Bedrock, Step Functions, Glue, EMR, Redshift, Athena, Docker, CI/CD
- **Systems:** Observability, Access Control, Data Quality, A/B Experimentation, Search Systems, Enterprise Integration
- **Languages:** English, Japanese (JLPT N1), Mandarin Chinese

EXPERIENCE

- **Amazon (Japan Consumer Innovation)** Tokyo, Japan
Machine Learning Engineer July 2025 – Present
 - Architected and operate JP Retail MCP, an AI analytics platform that lets retail teams query 115+ metrics, knowledge, workflows, and science model outputs through natural language; grew adoption to 301 monthly active users and 24K invocations in May 2026.
 - Built the platform's online query path, offline ETL path, domain skill warehouse, IAM-authenticated MCP gateway, SQL validation, caller attribution, structured logs, and CloudWatch metrics for 6+ departments.
 - Designed an agent evaluation framework with deterministic graders, SQL and tool-call validation, partial-credit checks, and LLM-as-Judge scoring; won two Japan Store Hackathon awards and made it the release gate for new agent capabilities.
 - Built a causal inference ML framework on AWS and SageMaker, covering feature-store construction, estimator comparison with EconML, and production training pipelines; reduced model iteration from weeks to hours.
 - Built a reusable AWS CDK data-access platform for Spark-backed pipelines and ad-hoc analysis, standardizing Redshift, Glue, EMR, IAM, and Lake Formation provisioning; onboarded 8 teams and cut setup from weeks to 1 day.
 - Mentor junior engineers and business intelligence engineers through project planning, technical design, implementation reviews, and career development.
- **Amazon (Product Search)** Tokyo, Japan
Software Development Engineer Apr 2023 – June 2025
 - Designed and built an offline evaluation system for search refinement selection, using traffic replay, feature flags, paired statistical analysis, relevance scoring, duplication analysis, and daily monitoring for rapid pre-A/B quality signals.
 - Turned the evaluation system into a production guardrail for search navigation data quality, replacing slow weblab-only feedback loops with automated checks and reducing algorithm iteration from months to hours.

EDUCATION

- **Tokyo Institute of Technology** Tokyo, Japan
M.Eng. in Information and Communication Engineering Sep 2020 – Sep 2022
- **Huazhong University of Science and Technology** Wuhan, China
B.Eng. in Electronic Information Engineering Sep 2016 – Jun 2020